

1. DESIGN.MD

Markdown

DESIGN SPECIFICATION — Aarush Karak Portfolio

1. Aesthetic Vision & Theme

- **Theme Name:** Cyber Dark Glassmorphism
- **Base Background Color:** `#0d1117`
- **Primary Color Palette:**
 - Electric Cyan: `#00f2fe`
 - Neon Crimson: `#e60049`
 - Soft Coral / Accent: `#d67964`
 - Deep Dark Base: `#0d1117`
- **Visual Style:** Modern, interactive, dark-mode glassmorphic cards over dynamic procedural webgl canvas backgrounds. High contrast typography with clean layout scannability.

2. Global Runtime Script Requirement

Include the dynamic effect runtime in `app/layout.tsx`:

```
```tsx
```

```
import Script from "next/script";
```

```
export default function RootLayout({ children }: { children: React.ReactNode }) {
 return (
 <html lang="en" className="dark">
 <body className="bg-[#0d1117] text-white antialiased min-h-screen">
 {children}
 <Script
src="https://cdn.aidesigner.ai/effects/runtime/v1.js" strategy="afterInteractive"/>
 </body>
 </html>
);
}
```

## 3. Dynamic Page Background Configurations

Every page route must be wrapped in a container with `className="relative isolate min-h-screen overflow-hidden"` to keep the canvas effect pinned strictly behind content (`-z-10`).

**Page 1: Home Page (/)**

TypeScript

```
<div
 data-aifx="fluid-smoke"
 data-aifx-colors="#00f2fe,#e60049,#00f2fe,#00f2fe"
 data-aifx-bg="#0d1117"
 data-aifx-bg-alpha="0.47"
 data-aifx-speed="0.64"
 data-aifx-scale="1.87"
 data-aifx-warp="1.53"
 data-aifx-rise="0.57"
 data-aifx-swirl="0.55"
 data-aifx-contrast="1"
 data-aifx-softness="0.78"
 className="absolute inset-0 -z-10 pointer-events-none"
 aria-hidden="true"
/>
```

## Page 2: About Page (/about)

TypeScript

```
<div
 data-aifx="fluted-glass"
 data-aifx-colors="#e60049,#00f2fe,#0d1117,#0d1117"
 data-aifx-bg="#0d1117"
 data-aifx-speed="0.79"
 data-aifx-flutes="39"
 data-aifx-shape="pattern"
 data-aifx-profile="contour"
 data-aifx-distortion="0.8"
 data-aifx-highlight="2"
 data-aifx-blur="2.5"
 data-aifx-angle="251"
 data-aifx-grain="0"
 data-aifx-mouse="1"
 className="absolute inset-0 -z-10 pointer-events-none"
 aria-hidden="true"
/>
```

## Page 3: Experience Page (/experience)

TypeScript

```
<div
 data-aifx="liquid-metal"
 data-aifx-colors="#e60049,#e60049,#00f2fe,#00f2fe"
 data-aifx-bg="#0d1117"
 data-aifx-speed="1.78"
```

```
data-aifx-scale="1.58"
data-aifx-contrast="2.2"
data-aifx-flow-angle="33"
data-aifx-relief="2.05"
data-aifx-mouse="0.5"
className="absolute inset-0 -z-10 pointer-events-none"
aria-hidden="true"
/>
```

## Page 4: Skills Page (/skills)

TypeScript

```
<div
 data-aifx="noise-shimmer"
 data-aifx-colors="#00f2fe,#e60049,#d67964"
 data-aifx-bg="#0d1117"
 data-aifx-scale="0.7"
 data-aifx-shimmer="1"
 data-aifx-intensity="0.69"
 data-aifx-contrast="2.5"
 data-aifx-speed="2"
 className="absolute inset-0 -z-10 pointer-events-none"
 aria-hidden="true"
/>
```

## Page 5: Blog Page (/blog)

TypeScript

```
<div
 data-aifx="aurora-flow"
 data-aifx-colors="#00f2fe,#e60049,#d67964"
 data-aifx-bg="#0d1117"
 className="absolute inset-0 -z-10 pointer-events-none"
 aria-hidden="true"
/>
```

## Page 6: Contact Page (/contact)

TypeScript

```
<div
 data-aifx="mesh-gradient"
 data-aifx-colors="#e60049,#e60049,#00f2fe,#0d1117"
 data-aifx-bg="#0d1117"
 data-aifx-speed="1.04"
 data-aifx-scale="2.09"
 data-aifx-warp="1.5"
```

```

data-aifx-softness="1"
data-aifx-contrast="1.16"
data-aifx-grain="0.2"
data-aifx-mouse="1"
className="absolute inset-0 -z-10 pointer-events-none"
aria-hidden="true"
/>

```

## 4. Component Glassmorphism Styling

All card elements throughout the application must inherit or utilize the `.glass-card` styling specification. Adjust layout dimensions (`w-full`, `h-auto`, grid systems) dynamically while maintaining core glass attributes.

### CSS Global Class Specification

Add the following to `globals.css`:

CSS

```

@layer utilities {
 .glass-card {
 background: rgba(255, 255, 255, 0.08);
 backdrop-filter: blur(50px);
 -webkit-backdrop-filter: blur(50px);
 border-radius: 20px;
 border: 1px solid rgba(255, 255, 255, 0.18);
 box-shadow:
 0 8px 32px rgba(0, 0, 0, 0.25),
 inset 0 1px 0 rgba(255, 255, 255, 0.3),
 inset 0 -1px 0 rgba(255, 255, 255, 0.1);
 position: relative;
 overflow: hidden;
 transition: transform 0.3s ease, border-color 0.3s ease, box-shadow 0.3s ease;
 }

 .glass-card:hover {
 transform: translateY(-4px);
 border-color: rgba(0, 242, 254, 0.4);
 box-shadow:
 0 12px 40px rgba(0, 242, 254, 0.15),
 inset 0 1px 0 rgba(255, 255, 255, 0.5);
 }

 .glass-card::before {
 content: "";
 position: absolute;

```

```

top: 0;
left: 0;
right: 0;
height: 1px;
background: linear-gradient(
 90deg,
 transparent,
 rgba(255, 255, 255, 0.8),
 transparent
);
pointer-events: none;
}

.glass-card::after {
 content: "";
 position: absolute;
 top: 0;
 left: 0;
 width: 1px;
 height: 100%;
 background: linear-gradient(
 180deg,
 rgba(255, 255, 255, 0.8),
 transparent,
 rgba(255, 255, 255, 0.3)
);
 pointer-events: none;
}
}

```

---

## ## 2. PRD.MD

```markdown

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Title

****Aarush Karak — Personal Portfolio Website****

Core Objective

Engineered as a high-performance modern web platform presenting the portfolio, software engineering projects, work history, skills, certifications, blog, and leadership achievements of Aarush Karak. The platform leverages Next.js App Router, Tailwind CSS, high-tech glassmorphism visual components, and dynamic GPU shader canvas backgrounds.

Target Profile Summary

- **Name:** Aarush Karak
- **Role:** Founder @ TheCoderBros | Software Developer | Spatial Computing & AI Specialist | Student @ John Fraser Secondary School
- **Location:** Greater Toronto Area, Ontario, Canada
- **Email:** worldshaans@gmail.com / 1135807@pdsb.net
- **Phone:** +1 (647) 519-4798
- **Live Portfolio Link:** <https://aarushkarak.vercel.app/>
- **LinkedIn:** <https://www.linkedin.com/in/aarush-karak/>

Navigation & Architecture

The application consists of 6 core pages accessible via a sticky dynamic glassmorphic header navigation bar with route indicator highlights and mobile responsive drawer menu:

1. **Home (/)** — Hero presentation, key stats, featured projects teaser, quick CTA.
2. **About (/about)** — Bio, background, education history, extracurricular leadership, references.
3. **Experience (/experience)** — Full chronological engineering work experience, open-source work, internships, and leadership roles.
4. **Skills (/skills)** — Interactive catalog of programming languages, tools, frameworks, and verified certifications.
5. **Blog (/blog)** — Technical writeups, devlogs, project case studies, and engineering updates.
6. **Contact (/contact)** — Functional interactive contact card, direct social links, reference contacts, and resume download link.

Detailed Page Content Specifications

1. Home Page (/)

- **Background Effect:** `fluid-smoke`
- **Hero Section:**
 - **Headline:** "Aarush Karak"
 - **Sub-Headline:** "Sophomore @ John Fraser Secondary School | Founder @ TheCoderBros | Software Developer | Spatial Computing & AI Enthusiast"
 - **High-impact summary badge grid:**
 - 🚀 Founder @ TheCoderBros
 - 🛠️ Platform Engineer @ Hack Club (Stardance Challenge)
 - 🏆 1st Place TSA State Champion (Video Game 3D Design)
 - 🎓 NYLF Engineering Invited Scholar
 - **Call To Actions:** `[Explore Projects]` (scrolls to featured) & `[Download Resume]` (links to PDF download)
- **Featured Software Showcase (Glass Cards):**

- **HELIOS:** Browser-based AI operating system with touch-free hand-gesture navigation and a 3D spatial interface. Built with Three.js & MediaPipe.
- **astro-tasks:** Python CLI developer dashboard on PyPI integrating GitHub CLI, git health checks, and stats for the Hack Club Stardance Challenge.
- **Finance Hub:** Full-stack financial intelligence platform with live market monitoring and AI-driven prediction exploration.
- **Knowledge-Globe:** Interactive 3D encyclopedia visualising knowledge networks on a 3D interactive globe.
- **FixMate:** AI-powered home maintenance diagnostic application.

2. About Page (`/about`)

- **Background Effect:** `fluted-glass`
- **Biography Card:**
 - Grade 10 student passionate about computer science, software development, AI systems, and spatial computing.
 - Builds end-to-end applications utilizing Python, JavaScript, Java, C++, Three.js, MediaPipe, REST APIs, PostgreSQL, React, and Node.js.
 - Selected through a competitive interview process as the youngest Maker Studio Teen Intern at Sci-Tech Discovery Center (Frisco, TX).
 - Former Elected Parliamentarian for TSA (guided 200+ students, 30+ state finalists).
- **Education Section (Glass Grid):**
 - **John Fraser Secondary School** — Mississauga, ON (Feb 2026 – Present)
 - **Coppell High School** — Coppell, TX (Aug 2025 – Jan 2026)
 - **Coppell Middle School West** — Coppell, TX (Aug 2024 – May 2025)
 - **Russian School of Mathematics** — Mathematics (2024 – 2029)
 - **Garden High School** — India (2014 – 2018)
- **Extracurriculars & Hobbies Card:**
 - **Pianist:** Classical and modern pianist (2015 – Present). Received Medals of Distinction in Grade 1 & Grade 2 Trinity College examinations.
 - **Badminton Player:** Competitive player (2022 – Present). Competed in 2 USA Badminton Open Regional Championships and 3 Open Local Championships.
 - **Sports:** Soccer, Tennis.
- **Volunteering & Community:**
 - **National Junior Honor Society (NJHS):** 50+ hours volunteer service, raised \$3,000+ for the Dallas Children's Advocacy Center.
 - **Ankur DFW:** 200 hours volunteer service across blood drives, community cleanups, food & clothing drives.

3. Experience Page (`/experience`)

- **Background Effect:** `liquid-metal`
- **Timeline Cards (Glassmorphism Styled):**
 - **TheCoderBros — Co-Founder & CEO** (July 2023 – Present)*
 - Founded student-led tech initiative; architected open-source AI, 3D web, and spatial computing applications.

- **Hack Club — Stardance Challenge Platform Engineer** (June 2026 – Present)*
 - Built and published `astro-tasks` PyPI package (Python CLI dashboard) in partnership with NASA, GitHub, and AMD. Logged 14 devlogs.
- **Hack Club — Hackatime Coder** (May 2026 – Present)*
 - Tracked 200+ hours of activity across 7 projects; maintained a 100% heartbeat acceptance rate via automated 24/7 pipelines.
- **Hack Club — OAuth Application Developer** (June 2025 – Present)*
 - Developed full-stack mobile/web app platform with serverless reactive architecture and optimized real-time sync.
- **CommunityOne — Back End Developer** (June 2026 – Present)*
 - Built cloud-native backend systems for open-source civic data platform centralizing public records and legislative datasets.
- **Chingu — Full-Stack SWE** (June 2026 – Present)*
 - Participated in 6-week Agile Voyage cohort building React/Node.js/Python/PostgreSQL applications with automated data pipelines.
- **Fiverr — Web Developer & Freelancer** (April 2025 – Present)*
 - Delivered over \$5,000 in high-value client projects specializing in 3D web environments, AI apps, and financial tools.
- **Technology Student Association (TSA) — Parliamentarian** (Aug 2025 – May 2026)*
 - Officer for 9th Grade Center (200+ students, 30+ state finalists). Awarded **1st Place in Video Game 3D Design** at TSA State Conference.
- **Sci-Tech Discovery Center — Maker Studio Teen Intern** (June 2025 – Sept 2025)*
 - Completed 40+ hours in STEAM makerspace teaching technical concepts. Youngest intern selected through competitive interview process.

4. Skills Page (`/skills`)

- **Background Effect:** `noise-shimmer`
- **Technical Skills Categorized Cards:**
 - **Languages:** Python, JavaScript, TypeScript, Java, C++, HTML5/CSS3, SQL
 - **Frameworks & Libraries:** React, Next.js, Node.js, Three.js, MediaPipe, REST APIs, PostgreSQL
 - **Domains:** Spatial Computing, AI Systems, UI/UX Design, Web Development, CLI Development
 - **Spoken Languages:** English (Native), Hindi (Native), Bengali (Native), Spanish (Limited), Marathi (Limited)
 - **Soft Skills & Leadership:** Public Speaking, Youth Mentorship, Agile Methodologies, Parliamentary Procedure
- **Verified Certifications Grid:**
 - Artificial Intelligence Fundamentals — IBM (June 2026)
 - Introduction to Generative AI — Google (June 2026)
 - Career Essentials in GitHub — GitHub (June 2026)
 - Intro to Machine Learning — Kaggle (April 2026)
 - Advanced SQL — Kaggle
 - AI Fluency Framework & Foundations
 - Claude Code in Action
 - OpenCV Bootcamp

- **Honors & Awards Grid:**

- 🏆 First Place: Video Game 3D Design — TSA State Conference (May 2026)
- 🎓 Invited Scholar — National Youth Leadership Forum (NYLF): Engineering (George Mason University)
- 🏠 Abacus Grand Master — SIP Academy India & International (Jan 2022)
- 🏆 Certificate of Distinction (Gold Medal, Rank #8) — SOF International Math Olympiad (Jan 2021)
- 🏆 Medals of Distinction (Grade 1 & 2) — Trinity College / University of West London

5. Blog Page (`/blog`)

- **Background Effect:** `aurora-flow`
- **Devlog & Technical Articles Grid (Glass Cards):**
 - **Building `astro-tasks`: Python CLI Developer Dashboard for NASA x Hack Club Stundance** (Published July 2026)
 - How I designed, packaged, and published a modular CLI to PyPI integrating GitHub notifications and git health checks.
 - **Spatial Computing on the Web: MediaPipe + Three.js** (Published June 2026)
 - Lessons learned from building HELIOS—a gesture-controlled browser OS without hardware wearables.
 - **Tracking 200+ Hours with Hackatime & Automated Pipelines** (Published May 2026)
 - Setting up 24/7 automated pipeline infrastructure for 100% heartbeat acceptance.
 - **Civic Tech and Open Source Data Platforms** (Published June 2026)
 - Architecting backend tools for CommunityOne to bridge local public record accessibility gaps.

6. Contact Page (`/contact`)

- **Background Effect:** `mesh-gradient`
- **Interactive Contact Form Card:**
 - Fields: Name, Email, Subject, Message, Submit Button (`.glass-card` styled with active hover gradients).
- **Direct Contact Information Card:**
 - Email: `worldshaans@gmail.com` | `1135807@pdsb.net`
 - Phone: `+1 (647) 519-4798`
 - Location: Greater Toronto Area, Ontario, Canada
 - LinkedIn: `https://www.linkedin.com/in/aarush-karak`
 - GitHub / Web: `https://aarushkarak.vercel.app/`
- **Professional References Card:**
 - **Jacie Hood** — Director of Education, Sci-Tech Discovery Center | 469-896-2188 | `jacie.hood@scitechdiscovery.org`
 - **Abby Resendez** — Media & Computer Science Teacher, Coppell ISD | `aresendez@coppellisd.com`